

# Database Structure Proposal

Team 23 · The Insights Lab | CSN Patient-Navigator Foundation

## Working assumption & design principles

- **Working assumption — row grain:** one row per (patient\_question × source) pair. In other words, each row is one website's answer to one patient question. So if kanker.nl and IKNL both answer 'How long do I wait for treatment?', that's two rows — one per source. For cancer-specific questions, we get one row per cancer type, so a single question can produce up to ~20 rows. Also see open question #1 and "What it looks like in practice" sections below.
- **Auditability first** — every row carries source URL, scrape timestamp, and a content hash.
- **Version-friendly** — re-scrapes do not overwrite; we append and flip is\_latest to preserve history.
- **Quality over coverage** — not all 6 sources will hit every column; missing values are explicit, not silent.

## Proposed columns for csn\_knowledge (single table)

| #  | Column           | Type   | Purpose & rule                                                       | Example                                        |
|----|------------------|--------|----------------------------------------------------------------------|------------------------------------------------|
| 1  | row_id           | INT PK | Surrogate key                                                        | 10247                                          |
| 2  | source_name      | TEXT   | Origin website (one of the 6 approved sources)                       | iknl.nl                                        |
| 3  | source_url       | TEXT   | Exact page scraped                                                   | iknl.nl/kankersoorten/borstkanker              |
| 4  | trustworthiness  | REAL   | Score from Makhotsa's matrix (0–5)                                   | 3.80                                           |
| 5  | cancer_type      | TEXT   | Cancer entity (NULL if generic)                                      | Borstkanker                                    |
| 6  | topic            | TEXT   | Primary CSN care-pathway topic; alternates go in notes               | Waiting Times                                  |
| 7  | journey_stage    | TEXT   | Primary stage: Pre-Dx / Dx / Tx / Post-Tx / Support                  | Diagnosis                                      |
| 8  | patient_question | TEXT   | The question this block answers                                      | How long until I start treatment?              |
| 9  | question_id      | INT FK | ID into CSN_question_bank_v1.xlsx (32 questions, 6 topics; see Q#1)  | Q-014                                          |
| 10 | english_summary  | TEXT   | Human-written chatbot-ready answer (≤ 60 words) — not auto-generated | Typically 2–6 weeks...                         |
| 11 | english_full     | TEXT   | Full translated paragraph (machine + Dutch-speaker QA)               | After diagnosis, the multidisciplinary team... |
| 12 | original_dutch   | TEXT   | Raw Dutch text (for QA & re-translation)                             | Na de diagnose...                              |
| 13 | content_hash     | TEXT   | SHA-256 of original_dutch — detects upstream change                  | a3f1...9c2b                                    |
| 14 | scraped_at       | DATE   | When this block was captured                                         | 2026-05-26                                     |
| 15 | is_latest        | BOOL   | TRUE = current canonical row for this hash                           | TRUE                                           |
| 16 | translator       | TEXT   | Tool/person who translated                                           | deep-translator/google                         |
| 17 | validated_by     | TEXT   | Dutch-speaker who QA'd (or NULL) — see Q#3                           | L. Khankishiyeva                               |
| 18 | notes            | TEXT   | Caveats; alternate topics/stages; edge cases                         | Also spans 'Communication'                     |

Shaded rows are the columns Harry specifically requested in the facilitator session (patient-question mapping, content hash, is\_latest flag).

## How the chatbot uses one table

```
SELECT english_summary, source_url, trustworthiness
FROM   csn_knowledge
WHERE  question_id = 'Q-014' AND is_latest = TRUE
ORDER BY trustworthiness DESC LIMIT 1;
```

Returns the best-scoring current answer with its citation in one join-free query. (

## Open questions for the team (review before Thursday 4:30 PM)

- ✓ **Q#1 — Question bank & row grain.** Working assumption: one row per (patient\_question × source). Companion file CSN\_question\_bank\_v1.xlsx contains 32 draft questions across CSN's 6 topics<sup>1</sup>. Do we (a) freeze at 32, (b) expand via CSN community FAQs, or (c) trim based on what scraped content actually covers?
- ✓ **Q#2 — Controlled vocabularies.** Freeze topic at CSN's 6 + 'Other' and journey\_stage at the 5 above, before tagging begins?
- ✓ **Q#3 — Validation & ranking.** Dutch-speaker QA on all rows or only chatbot-surfaced rows? And when sources tie, rank by trustworthiness alone, or also by recency / validation status?

<sup>1</sup> These are a draft I generated by reasoning from CSN's 6 care-pathway topics and the 5 patient-journey stages. I would like to ask Makhotsa to validate them against real sources before locking them in.

## What it looks like in practice

Let's take one real patient question from the bank — **Q-008: "How long until I start treatment after diagnosis?"** This question can be answered by *multiple* websites. So in the database, it produces **multiple rows** — one per website that has an answer.

| row_id | source_name                  | question_id | patient_question                                    | english_summary                                                                 | trustworthiness |
|--------|------------------------------|-------------|-----------------------------------------------------|---------------------------------------------------------------------------------|-----------------|
| 1247   | <a href="#">iknl.nl</a>      | Q-008       | "How long until I start treatment after diagnosis?" | "In the Netherlands, treatment typically begins within 3 weeks of diagnosis..." | 3.80            |
| 1248   | <a href="#">kanker.nl</a>    | Q-008       | "How long until I start treatment after diagnosis?" | "Usually 2–6 weeks; depends on cancer type and treatment plan..."               | 4.90            |
| 1249   | <a href="#">thuisarts.nl</a> | Q-008       | "How long until I start treatment after diagnosis?" | "Your GP and specialist will agree a timeline, usually within a month..."       | 4.80            |

**Three rows, same question, three different sources, three different answers.** The chatbot will rank them by trustworthiness, for example, and pick the best one.

## What about IKNL data with 23 cancer types?

This is where it gets a little more interesting. Some patient questions are **cancer-specific**:

- ✓ "What are the treatment options for breast cancer?" (specific to one cancer)
- ✓ "How long until I start treatment?" (general — applies to all cancers)

For cancer-specific questions, **one question + one source can produce multiple rows** — one per cancer type. Like this:

| row_id | source                  | question_id | cancer_type          | patient_question                    | english_summary                              |
|--------|-------------------------|-------------|----------------------|-------------------------------------|----------------------------------------------|
| 2001   | <a href="#">iknl.nl</a> | Q-011       | Borstkanker (breast) | "What treatment options do I have?" | "For breast cancer: surgery, chemo..."       |
| 2002   | <a href="#">iknl.nl</a> | Q-011       | Longkanker (lung)    | "What treatment options do I have?" | "For lung cancer: surgery, immunotherapy..." |
| 2003   | <a href="#">iknl.nl</a> | Q-011       | Darmkanker (colon)   | "What treatment options do I have?" | "For colon cancer: surgery, radiotherapy..." |

So Q-011 alone, just from IKNL, could produce up to **23 rows** (one per cancer type IKNL covers).